

Clinical Reference Guide: Prostate Cancer

Extracted Focus & Question-Answer Summary from Cancer.gov PDQ

DOCUMENT OVERVIEW

Focus Condition:	Prostate Cancer [cite: 21, 22, 23]
Semantic Group:	Disorders (T191) [cite: 21, 22, 23]
Source Data:	National Cancer Institute (NCI) / CancerGov [cite: 21, 22, 23]

Question 1: What is (are) Prostate Cancer ?

- Prostate cancer is a disease in which malignant (cancer) cells form in the tissues of the prostate [cite: 21, 22, 23].
- The prostate is a gland in the male reproductive system located just below the bladder and in front of the rectum [cite: 21, 22, 23].
- It is about the size of a walnut and surrounds part of the urethra, producing fluid that makes up part of the semen [cite: 21, 22, 23].
- Prostate cancer is the most common nonskin cancer (or second most common overall) among men in the United States [cite: 22, 23].
- About 1 out of 5 men will be diagnosed with it, though most men diagnosed do not die from it [cite: 21, 22, 23].
- As men age, the prostate may enlarge causing benign prostatic hyperplasia (BPH), which blocks urine flow and causes similar symptoms but is not cancer [cite: 21, 22, 23].

Question 2: Who is at risk and what are protective factors?

Risk Factors:

- **Age:** Prostate cancer is rare in men under 50, and the chance of developing it increases with age [cite: 22].
- **Race:** It occurs more often, and with higher mortality, in African-American men than in white men [cite: 22, 23].
- **Family History:** Men with a father, brother, or son who had prostate cancer have a higher-than-average risk [cite: 22].
- **Hormones:** Dihydrotestosterone (DHT) can cause the prostate to enlarge and may play a role in cancer development [cite: 22].
- **Supplements & Diet:** Taking Vitamin E alone or 1 milligram supplements of folic acid increases risk [cite: 22]. A diet high in dairy and calcium may also cause a small increase in risk [cite: 22].

Protective Factors:

- **Folate:** Adequate folate naturally present in the diet lowers risk [cite: 22].
- **Medications:** Finasteride and dutasteride block the enzyme that changes testosterone to DHT, lowering the overall risk of prostate cancer (though they may increase the incidence of more aggressive tumors and cause side effects like erectile dysfunction) [cite: 22].

Factors With Unclear/No Effect:

- Selenium, lycopene, general diet (fat/fruits/vegetables), and regular multivitamin use have not been proven to affect risk, though excessive multivitamin use may increase the risk of advanced prostate cancer [cite: 22].

Question 3: What are the symptoms of Prostate Cancer ?

- Weak or interrupted ("stop-and-go") flow of urine, or a sudden urge to urinate [cite: 21].
- Frequent urination, especially at night [cite: 21].
- Trouble starting the flow of urine or emptying the bladder completely [cite: 21].
- Pain or burning while urinating [cite: 21].
- Blood in the urine or semen [cite: 21].
- A pain in the back, hips, or pelvis that doesn't go away [cite: 21].
- Shortness of breath, feeling very tired, fast heartbeat, dizziness, or pale skin caused by anemia [cite: 21].

Question 4: How to diagnose Prostate Cancer ?

- **Physical exam and history:** Checking general health and past illnesses [cite: 21].
- **Digital rectal exam (DRE):** A doctor inserts a lubricated, gloved finger into the rectum to feel the prostate for lumps [cite: 21].
- **Prostate-specific antigen (PSA) test:** Measures PSA levels in the blood, which may be elevated in men with prostate cancer, infection, or BPH [cite: 21].
- **Transrectal ultrasound & MRI:** Inserting a probe into the rectum to bounce sound waves or use magnets to create detailed pictures of the prostate and nearby tissues [cite: 21].
- **Biopsy:** Removal of tissue via a thin needle through the rectum to check for cancer cells under a microscope [cite: 21]. The pathologist determines the Gleason score (ranging from 2-10), indicating how likely the tumor is to spread [cite: 21].

Question 5: What is the outlook for Prostate Cancer ?

- Prognosis and treatment options depend on the stage of the cancer, including the PSA level, Gleason score, tumor grade, and extent of spread [cite: 21].

- The patient's age, general health, past treatments, expected side effects, and personal wishes also affect treatment choices [cite: 21].
- Most men diagnosed with prostate cancer do not die of it [cite: 21, 22, 23].

Question 6: What are the stages of Prostate Cancer ?

Staging utilizes tests like bone scans, MRI, CT scans, pelvic lymphadenectomy, seminal vesicle biopsy, and ProstaScint scans [cite: 21].

- **Stage I:** Cancer is found in the prostate only, usually with a PSA lower than 10 and a Gleason score of 6 or lower [cite: 21].
- **Stage II:** Cancer is more advanced but still confined to the prostate (divided into IIA and IIB based on PSA levels up to 20 or higher, and Gleason scores up to 10) [cite: 21].
- **Stage III:** Cancer has spread beyond the outer layer of the prostate and may involve the seminal vesicles [cite: 21].
- **Stage IV:** Cancer has spread to nearby tissue/organs (rectum, bladder, pelvic wall), lymph nodes, or distant parts of the body such as bones [cite: 21].

Question 7: What are the treatments for Prostate Cancer ?

- **Watchful Waiting / Active Surveillance:** Closely monitoring the condition without active treatment until symptoms appear or test results change [cite: 21].
- **Surgery:** Radical prostatectomy (retropubic or perineal) to remove the prostate, and pelvic lymphadenectomy [cite: 21]. Transurethral resection of the prostate (TURP) may relieve symptoms [cite: 21]. Surgery can cause impotence, urinary/stool leakage, penile shortening, or inguinal hernia [cite: 21].
- **Radiation & Radiopharmaceutical Therapy:** External radiation (including conformal and hypofractionated), internal radiation (radioactive seeds), and alpha emitter radiation (Radium-223) for bone metastases [cite: 21].
- **Hormone Therapy:** Treatments that remove or block male hormones (testosterone/DHT) using LHRH agonists, antiandrogens, adrenal suppressants, orchiectomy, or estrogens [cite: 21].
- **Chemotherapy & Biologic Therapy:** Systemic or regional drugs to stop cell division, or biologic therapies (like Sipuleucel-T) to boost the immune system [cite: 21].
- **Bisphosphonate Therapy:** Drugs like clodronate or zoledronate to reduce bone disease and fracture risk [cite: 21].
- **Targeted/New Therapies:** Cryosurgery, high-intensity focused ultrasound, and proton beam radiation are being tested [cite: 21]. Denosumab may be used to prevent bone metastases [cite: 21].

Question 8: What research (or clinical trials) is being done for Prostate Cancer ?

- Clinical trials study ways to lower the risk of developing prostate cancer, test new treatments, and find ways to prevent recurrence or reduce side effects [cite: 21, 22].
- Trials test interventions like dietary changes, vitamins, finasteride/dutasteride, new surgical techniques, and emerging radiation therapies [cite: 21, 22].

This document was generated for educational and clinical reference purposes based on official patient treatment guidelines (PDQ) from the National Cancer Institute.